The Hong Kong University of Science and Technology

Department of Computer Science and Engineering

COMP 4421 (Fall 2026)

Assignment 1

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Instructions/Hints:

1. The assignment should be submitted to Canvas as a zip file named **<SID>.zip**, which MUST contain all required **source codes, results** and one **PDF file**.
2. Please submit your assignment before 11:59 pm, Oct. 7, 2026.
3. Late submission will incur a 20% penalty and no more acceptations after one day.
4. Unless otherwise specified, pixel coordinates are represented as **(row, column)**. Show intermediate calculations where appropriate.
5. If you have any questions, feel free to contact TAs.

Problem 1 Image Arithmetic and Histograms (10 pts)

Consider 5-bit grayscale images, where pixel intensities range from 0 to 31. Both images have size 40×50.

(a) Saturated image addition (5 pts)

Image A is a constant image in which every pixel has intensity 6.

The histogram of Image B is given below. All intensity levels not listed have zero pixels.

Intensity r	**0**	**2**	**5**	**10**	**15**	**20**	**25**	**27**	**29**	**31**
Count	100	150	250	300	300	300	250	150	100	100

A new image C is formed pixel-wise by

$$C(x, y) = \min(A(x, y) + B(x, y), 31).$$

Give all non-zero entries of the histogram of C. Clearly indicate the corresponding intensity levels and pixel counts.

B **intensity**	**Count**	$C = \min(B + 6, 31)$
0		
2		
5		
10		
15		
20		
25		
27		
29		
31		

(b) Difference imaging (5 pts)

Two registered images of the same scene are acquired at different times. Between the first and second acquisition:

- 150 pixels change from intensity 8 to intensity 13;
- 90 pixels change from intensity 18 to intensity 14;
- all remaining pixels are unchanged.

The signed difference image is defined as

$$D(x, y) = I_{\text{after}}(x, y) - I_{\text{before}}(x, y).$$

Give the histogram of D by listing all its non-zero bins.

Based on the difference image, state how many pixels became brighter, how many became darker, and how many remained unchanged.

Problem 2 Pixel Adjacency and Distance (15 pts)

Consider the following image subset:

$$\begin{bmatrix} 1 & 2 & 0 & 3 & 2 & 1 \\ 0 & 2 & 2 & 3 & 0 & 1 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 2 & 2 & 3 & 0 & 1 & 0 \\ 3 & 1 & 2 & 2 & 2 & 3 \\ 1 & 0 & 1 & 3 & 2 & 2 \end{bmatrix}.$$

Let

$$p = (3,6), \quad q = (6,5).$$

(a) $V = \{0, 1, 2\}$ (6 pts)

For pixels whose values belong to V:

1. Find one shortest **4-path** from p to q.
2. Find one shortest **8-path**.
3. Find one shortest **m-path**.

Give the length of each path.

(b) $V = \{2, 3\}$ (6 pts)

Repeat the analysis for $V = \{2, 3\}$.

For each adjacency definition, state whether a path exists. If one exists, give its shortest length. If no path exists, briefly explain why.

(c) Geometric distances (3 pts)

Ignoring pixel intensity values, let $r = (1, 1)$, and consider two pixels

$$a = (1, 6),\ b = (4, 4).$$

Calculate $D_4(r, a)$, $D_8(r, a)$, $D_4(r, b)$, and $D_8(r, b)$.

Which pixel is closer to r under D_4, and which pixel is closer under D_8? Briefly explain why the two distance measures may give different conclusions.

Problem 3 Spatial Filtering and Histogram Information (25 pts)

Consider two 10×10 binary images whose pixel values are either 0 or 1.

Image A: Columns 1–5 contain only 0s, while columns 6–10 contain only 1s.
Image B: Columns alternate between 0 and 1:

$$0,1,0,1,0,1,0,1,0,1.$$

Every row of Image B has this same pattern.

Therefore, both original images contain exactly 50 zero-valued pixels and 50 one-valued pixels.

A 3×3 averaging filter is defined as

$$h = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

The filter is applied with stride 1.

(a) Output dimensions and padding (5 pts)

Without padding, what is the size of the filtered output?

If the output is required to remain 10×10, determine the number of zero-padding rows or columns required on each side of the input: top, bottom, left, and right.

(b) Using valid filtering, i.e. without padding, will the filtered versions of Images A and B have the same histogram?

Answer YES or NO, and explain your reasoning without calculating every output pixel individually. **(5 pts)**

(c) For the 8×8 valid-filtering output, calculate all non-zero histogram entries for:

1. Filtered Image A;
2. Filtered Image B.

Possible output intensities are multiples of 1/9.

Your answer should specify both the output intensity and the corresponding number of pixels. **(10 pts)**

(d) Interpretation (5 pts)

In one or two sentences, explain why two images having identical original histograms does not imply that they will have identical histograms after spatial filtering.

Problem 4 Recovering an Affine Transformation (15 pts)

The affine transformation is

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}.$$

The three point correspondences are

$$(1,1) \rightarrow (4,3),$$
$$(3,1) \rightarrow (6,1),$$
$$(1,3) \rightarrow (6,5).$$

(a) Recover the transformation (8 pts)

Determine all six unknown parameters

$$a_{11}, a_{12}, a_{13}, a_{21}, a_{22}, a_{23}$$

and write the complete affine transformation matrix.

Show the system of equations used in your derivation.

(b) Transform a new point (4 pts)

Using your recovered matrix, calculate the transformed coordinates of $(2,4)$.

(c) Degenerate configuration (3 pts)

Explain why three source points lying on the same straight line are insufficient for uniquely recovering an arbitrary affine transformation, even though three point pairs normally provide six scalar equations.

Problem 5 Histogram Equalization (15 pts)

Consider a 3-bit grayscale image of size 30×20, so that there are 600 pixels and L=8.

Its intensity distribution is

r_k	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
n_k	30	60	90	150	120	90	40	20

Histogram equalization is defined as

$$s_k = \text{round}\left[(L-1)\sum_{j=0}^{k} p_r\left(r_j\right)\right].$$

(a) Equalization mapping (8 pts)

For every input intensity r_k, calculate:

$$p_r(r_k),$$

the cumulative distribution function,

$$CDF(r_k),$$

and the final mapped intensity

$$s_k.$$

(b) Equalized histogram (4 pts)

Determine the complete histogram of the equalized image.

(c) Interpretation (3 pts)

Is the resulting histogram perfectly uniform?

Explain why histogram equalization of a discrete digital image does **not necessarily** produce an exactly uniform histogram.

Also identify any different input intensity levels that are mapped to the same output intensity.

Programming Problem 6 (20 pts)

In this problem, you will implement Laplacian filtering and use bit-plane slicing to recover a hidden watermark. You are given the 8-bit grayscale image `watermarked_cameraman.png` and the MATLAB skeleton `problem6.m`.

Rules:

- Write code **only** at `% your code here`. **Modifying any other part of the skeleton gives 0 marks for that sub-question.**
- Do not use built-in filtering functions without permission (`imfilter`, `conv2`, `filter2`, `fspecial`, `del2`, etc.).
- Submit one zip file **<SID>.zip** containing `problem6.m`, the three generated images (`sharpened_K4.png`, `sharpened_K8.png`, `hidden_watermark.png`) and your written answers (PDF).
- Please carefully read the descriptions in the skeleton codes before coding.

(a) Use Laplacian kernels for edge extraction and sharpening (see Lecture 4, p. 70). Please pad the image border by zeros. **(12 pts)**

(i) (4 pts) Define `K4`, a Laplacian kernel. Filter `f` with it and store the result in `lap4`.
(ii) (4 pts) Define `K8`, the kernel that includes the diagonal terms. Filter `f` with it and store the result in `lap8`.
(iii) (2 pts) Sharpen the image by $g = f + c\nabla^2 f$ using `lap4`. Clip to [0, 255], convert to `uint8`, and store the result in `sharp4`. **Hint:** c = −1 if the kernel center is negative, and c = +1 if it is positive.
(iv) (2 pts) Repeat (iii) with `lap8` and store the result in `sharp8`.

(b) A binary watermark is hidden in one bit plane of the image. You can use `bitget()` in this question. **(8 pts)**

(i) (4 pts) First, slice the image into bitplanes and store the k-th bit plane (k = 1 is the least-significant bit) in `planes(:,:,k)`, scaled to 0/255.
(ii) (4 pts) Then, assign the bit plane containing the watermark to `watermark`.